

Gunakarthik Naidu Lanka

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Summary

Software engineer who completed an accelerated B.S. and M.S. in Computer Science back to back, with two years building production services in Python, Java, and SQL, from REST APIs and microservices to the databases and data pipelines underneath them. Comfortable owning a service end-to-end, from schema design and query optimization to deployment, monitoring, and the CI/CD pipeline that ships it.

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, SQL, Bash

Backend & APIs: FastAPI, Spring Boot, Node.js, REST APIs, GraphQL, Microservices, Event-Driven Architecture

Databases: PostgreSQL, MySQL, MongoDB, DynamoDB, Neo4j, SQLite

Data & Messaging: Apache Kafka, ZooKeeper, Kafka Connect, ETL/ELT Pipelines, Batch & Real-Time Streaming

Cloud & DevOps: AWS (EC2, S3, Lambda, RDS), Docker, Kubernetes, CI/CD, Git/GitHub, Linux

Practices: System Design, Distributed Systems, Code Reviews, Testing, Agile/Scrum

Professional Work Experience

IBU Solutions (Integrated Travel) | Data Analyst (AI Research)

Remote | January 2026 – Present

- Built and maintained a weekly Python data pipeline (ingestion, quantitative modeling, delivery) supporting a 6-paper research series spanning **14 cities** across the US, Canada, and India.
- Developed reproducible quantitative models with structured, well-documented workflows and clear technical reporting.

Velocity Tech | Software Engineering Intern

Remote | January 2024 – December 2024

- Developed backend microservices in Java Spring Boot and Python FastAPI, exposing 10+ REST endpoints, sustaining **99.9% uptime** in production.
- Optimized PostgreSQL schemas and ETL pipelines on **AWS RDS** processing **500K+ records**, reducing p99 retrieval latency **45%**.
- Containerized services with Docker and automated deployment through CI/CD, participating in code reviews and Agile sprints.

Arizona State University | Teaching Assistant, Data Structures & Algorithms

Tempe, AZ | January 2023 – December 2024

- Mentored **100+ students** across 4 semesters in Python, Java, and JavaScript, sustaining a **92% satisfaction** score.
- Led debugging workshops covering data structures, algorithms, and system design fundamentals.

EPICS at ASU | Full Stack Engineer Intern

Tempe, AZ | January 2022 – May 2022

- Built and deployed a full-stack application (React.js frontend, Python REST API backend, PostgreSQL) on AWS with Docker.
- Delivered 4 releases in 5 months with **95% automated test coverage**, collaborating across a cross-functional team.

Projects

HireAgent | Multi-service Python backend, production

- Built solo a 7-stage backend pipeline (FastAPI, SQLite state layer, LangGraph orchestration, multi-provider LLM routing) processing **700+ real cases** in production.
- Diagnosed a reliability bug and built a systematic evaluation harness that caught an entire class of failures, deployed with Docker and GitHub Actions CI/CD.

Real-Time Kafka Streaming Pipeline | Distributed ELT streaming system

- Built a production-grade real-time streaming pipeline on Kubernetes (ZooKeeper, Kafka broker, Kafka Connect, Neo4j Sink Connector) implementing CDC-style streaming, validated end-to-end.
- Deployed via Helm manifests on Kubernetes, designing the pipeline for horizontal scalability and fault tolerance.

Additional Projects & Open Source: Large-scale PostgreSQL ingestion (18M-row dataset, pg_bulkload, PL/pgSQL partitioning); NeighborhoodPulse (distributed backend aggregating multi-source data); open-source contribution to scikit-learn (PR #34685, merged, 60k+ star library).

Certifications

AI Infrastructure and Operations Fundamentals | NVIDIA

Credential ID LZ9QH003IM56

AI Infrastructure: Hypercomputer, Cloud GPUs & Cloud TPUs | Google Cloud

Credential ID K6LYXGBG3024

Agentic AI Certified Foundations Associate | Oracle

Credential ID 330419616AAI26OFA

AI Open Source Capstone Certificate – Honors | CodePath

Credential ID a304d57c-ff29-48a7-8d48-8263739e3a7e

Education

Arizona State University

Tempe, AZ | December 2025

M.S. in Computer Science | GPA 3.8/4.0

Arizona State University

Tempe, AZ | December 2024

B.S. in Computer Science, Minor in Business | GPA 4.0/4.0